PANDIT DEENDAYAL PETROLEUM UNIVERSITY, GANDHINAGAR
SCHOOL OF TECHNOLOGY

COURSE STRUCTURE FOR B. TECH. IN ELECTRICAL ENGINEERING

Semester I			B. Tech. in Electrical Engineering											
Sr. No.	Course/Lab Code	Course/Lab Name	Teaching Scheme					Examination Scheme						
			L	T	P	C	Hrs/Wk	Theory			Practical		Total
								MS	ES	IA	LW	LE/Viva	Marks
1	20MA101T	Mathematics – I	3	1	0	4	4	25	50	25	--	--	100
2	20CE101T	Elements of Civil Engineering and Solid Mechanics	4	0	0	4	4	25	50	25	-	-	100
3	20CE101P	Elements of Civil Engineering and Solid Mechanics Laboratory	0	0	2	1	2	--	--	--	25	25	50
4	20EE101T	Elements of Electrical Engineering	3	0	0	3	3	25	50	25	--	-	100
5	20EE101P	Elements of Electrical Engineering Laboratory	0	0	2	1	2	--	--	--	50	50	100
6	20PH101T	Engineering Physics	3	0	0	3	3	25	50	25	--	--	100
7	20PH101P	Engineering Physics Laboratory	0	0	2	1	2	--	--	--	50	50	100
8	16ME103P	Workshop Practice	0	0	2	1	2	--	--	--	25	25	50
9	20CP101T	Programming with C	1	0	0	1	1	25	50	25	--	--	100
10	20CP101P	Programming with C - Lab.	0	0	2	1	2	--	--	--	50	50	100
11	16HS109T	Professional Ethics & Human Values	1	0	0	1	1	25	50	25	--	--	100
12	16SP101/102/103	NCC-I/NSS-I/Sports-I	0	0	2	1	2	--	--	--	--	100	100
		Total	**15**	**1**	**12**	**22**	**26**	**150**	**300**	**150**	**200**	**300**	**1100**

IA- Internal Assessment, MS-Mid Semester; ES – End Semester Exam

Annexure 01

Teaching & Examination Scheme

B. Tech Electrical Engineering – 1st & 2nd Semester

Syllabus offered by Electrical Engineering in B. Tech 1st & 2nd Semester

Batch – EE2021

w.e.f. ACY 2021-2022

20EE101T					Elements of Electrical Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart knowledge on DC and AC circuits.
2. To learn construction, working principles and characteristics of transformer and induction machines.
3. To introduce students to various means for electrical safety and protection of electrical installations.
4. To impart knowledge on electric wiring and illumination for domestic and industrial applications.

UNIT I: DC CIRCUITS **10 Hrs**

Electrical circuit elements (R, L and C), voltage and current sources, dependent and independent sources, Ohms Law, temperature co-efficient of resistance, Kirchhoff current and voltage laws, voltage and current divider circuit, Thevenin and Norton Theorems and their equivalents, maximum power transfer and superposition theorems, nodal and mesh analysis, star-delta transformation, Time domain analysis/natural response of first order RL and RC Circuit

UNIT II: AC CIRCUITS **10 Hrs**

Generation of AC voltage, representation of sinusoidal waveforms, RMS values of different sinusoidal waveforms, Rectangular and Polar representation of phasor, Sinusoid representation in time and frequency domain, Analysis of single-phase ac series circuits consisting of R, L, C, RL, RC, RLC combinations, instantaneous power, average power and reactive power, complex power and power factor.AC parallel circuit and its solution in admittance form, resonance in AC series and parallel circuits, polyphase circuits, star and delta representation of polyphase circuits, power measurement in polyphase circuits.

UNIT III: TRANSFORMERS AND INDUCTION MACHINES **10 Hrs**

Magnetic material and its B-H characteristic, Faraday's Law of Electromagnetic Induction **Transformers:** ideal transformer, EMF equation for transformer, working of a practical transformer on no-load and load **Induction Machine:** Types of induction motors, production of rotating magnetic field from a 3-phase supply, operation of a 3-phase phase induction motor, starting and running torques, Torque-slip characteristics of induction motor, Power Stages inan induction motor.

UNIT IV: ELECTRICAL INSTALLATION, SAFETY AND PROTECTION **10 Hrs**

Fuse, MCB, ELCB, MCCB, underground cables, Domestic and Industrial Wiring, Types of lamps, illumination schemes and lumen requirements for domestic and industrial applications, Earthing and it schemes. Electrical safety rules, electric shock and first aid, energy conservation methods, elementary calculation of energy consumptions, tariffs.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Analyze electrical circuits using network theorems.
CO2 – Compare the behaviour of R, L and C and their combinations in AC circuits.
CO3 – Analyze balanced polyphase systems in star and delta configurations
CO4 – Understand the construction, working and basic characteristics of transformer and induction machines
CO5 – Recognize the importance of protective devices and electrical safety measures
CO6 – Perform domestic and industrial electrification

TEXT/REFERENCE BOOKS

1. J. Bird, **"Electrical Circuit Theory and Technology"**, Routledge, Tailor and Francis Group, Sixth Edition, 2017.
2. D. P. Kothari and I. J. Nagrath, **"Basic Electrical Engineering"**, Tata McGraw Hill, 2010.
3. B. L. Theraja, **"Electrical Technology",** Vol. 1, S. Chand Publication, New Delhi.
4. Surjit Singh, **"Electrical Estimating and Costing"**, Dhanpat Rai and Co.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks 60 Marks
Part B/Question: 4 Questions, one from each unit, each carrying 10 marks 40 Marks

20EE101P					Elements of Electrical Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	50

COURSE OBJECTIVES

1. To gain practical knowledge on DC and AC circuits
2. To learn operation of electrical instruments and electrical machines
3. To develop skills to implement electric wiring

LIST OF EXPERIMENTS

1. Introduction to elements of electrical engineering laboratory and to study different electrical measuring instruments
2. To validate Ohm's law with linear resistors and find power dissipation in resistor
3. To implement voltage divider and current divider circuit
4. To validate Thevenin and Norton theorem for DC circuit
5. To validate Superposition and Maximum Power Transfer theorem for DC circuit
6. To obtain transient response of RL and RC circuit
7. To evaluate performance of AC series circuit
8. To evaluate performance of AC parallel circuit
9. To analyse resonance condition in AC circuit
10. To establish relation between line and phase quantities in star and delta connected polyphase system
11. To measure power in polyphase system
12. To perform load test on 1-phase transformer
13. To perform load test on 3-phase transformer
14. To obtain current time characteristic for domestic protective devices
15. To carry out domestic electric wiring

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 – Operate basic electrical measuring instruments
CO2 – Simulate the basic electrical circuits and obtain results based on electrical laws and network theorem
CO3 – Understand the performance of AC circuit with different connection of R, L and C
CO4 – Formulate star and delta configuration of polyphase system and measure power in polyphase system
CO5 – Operate transformer and induction machines and evaluate its performance
CO6 – Understand the basic wiring and operation of protective devices for domestic application

END SEMESTER EXAMINATION PATTERN

Max. Marks: 100 **Exam Duration: 3Hrs**

Lab Work – Continuous Assessment 50 Marks
Lab Exam and Viva 50 Marks